

1. Overview of Computer Networks

- **Old paradigm: a single powerful computer serving all the needs of an organization**
- **New paradigm[†] - Computer networks: a large number of separate (autonomous) but internetworked (being able to exchange information) computers doing the job**
 - **Merging of computer and communications technologies – no geographical barrier**
 - **Connection: copper wire, fiber optics, microwaves, infrared, communication satellites, ...**

† Plus new applications: distributed processing (grid and cluster computing, pervasive computing, ...), WWW, ...

1.1 Uses of Computer Networks

a. Business[‡] applications

- for **resource sharing** including programs, equipment, data (mostly databases on central servers), ...
- a **communication medium** – e-mail, writing a report together by making changes on an online document
- videoconferencing – to hold meetings by hearing and seeing each other
- electronic business
 - business to business - placing orders, ...
 - business with consumers, usually called e-commerce – home shopping

[‡] Everything: government, commercial companies, ...

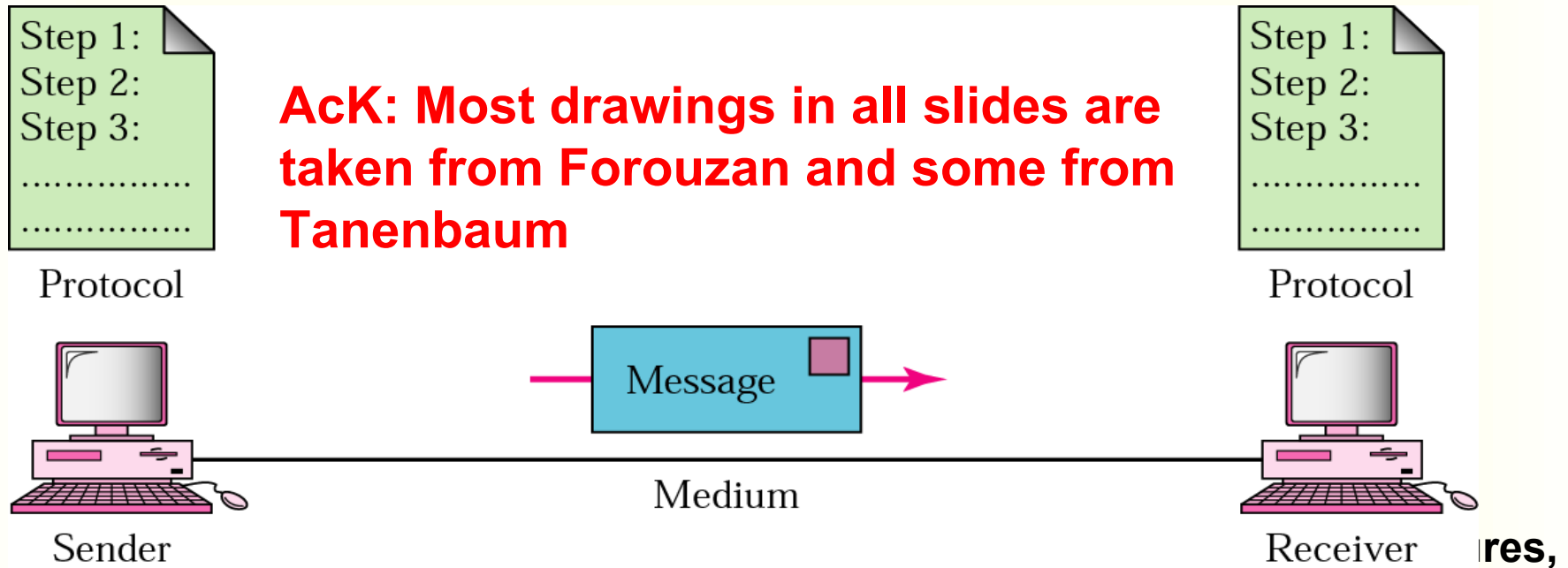
b. Home applications

- Access to remote information – newspapers, radio, on-line digital libraries (ACM, IEEE, ...), ...
- Person-to-person communication
 - **e-mail** (audio, video, pictures, ...)
 - **newsgroups** (not in real time)
 - **instant messaging** (between two people in real time, e.g., Yahoo Messenger, Skype),
 - **chat room** (for a group of people in real time)
 - using Internet to carry **telephone calls**, **video phone**, and **Internet radio**
- Interactive entertainment: video on demand, interactive television, games (virtual reality – with photographic-quality moving images)
- Electronic commerce – with online manuals

c. Mobile Users: using mobile computers - Laptop (notebook), Palmtop (PDAs), and handheld computers - and wireless networks in cars and **airplanes??**

1.2 Overview of Data Communications

- A data communication system has 5 components



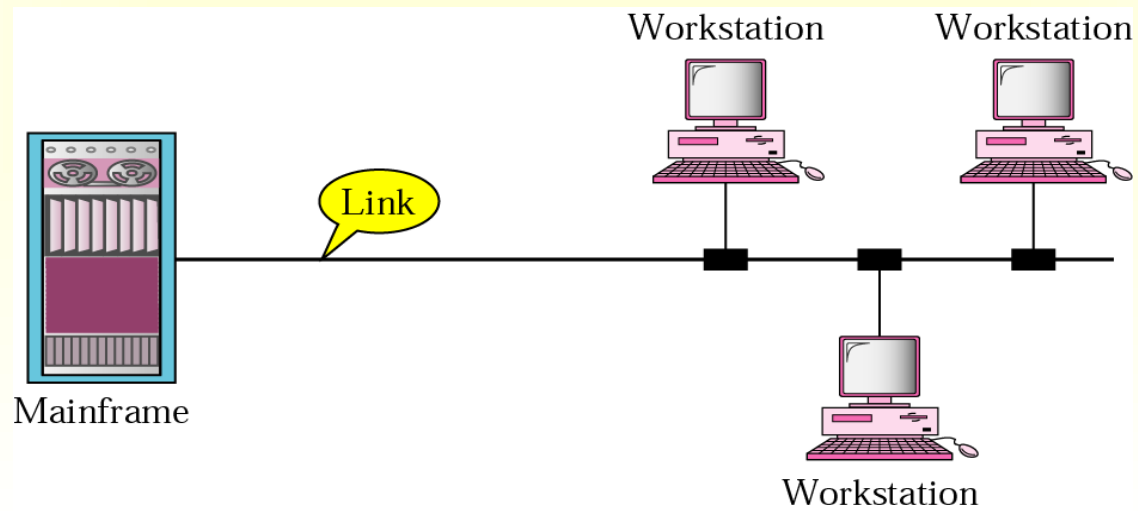
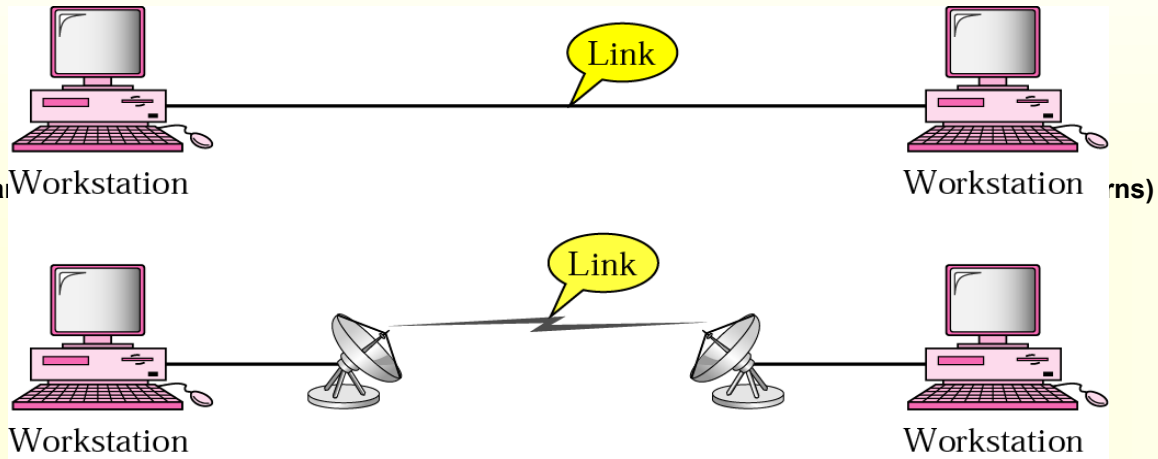
sound, video - or combinations)

2. **Sender:** the device - computer, video camera, ...
3. **Receiver:** still the device
4. **Medium:** the physical path by which a message travels from sender to receiver
5. **Protocol:** the set of rules that govern data communications; an agreement between the communicating devices

- **Direction of data flow** can be
 - **Simplex**: unidirectional, only one of the devices can transmit
 - **Half-duplex**: both can transmit and receive, but not at the same time
 - **Full-duplex**: both can transmit and receive at the same time

- **Network Criteria**
 - **Performance**: can be measured using **transit time** (amount of time required for a message to travel) and **response time** (elapsed time between an inquiry and the response)
 - **Reliability**: measured by the frequency of failure, the time it takes a link to recover from failure, and the network's robustness in a catastrophe
 - **Security**: protecting data from unauthorized access

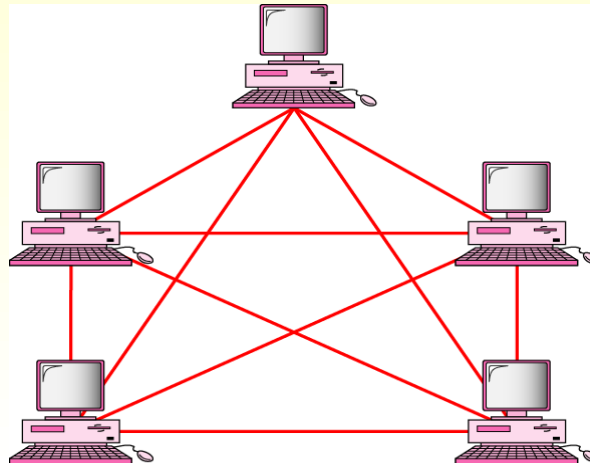
- Types of connection
 - Point-to-point:** provides a dedicated link between two devices



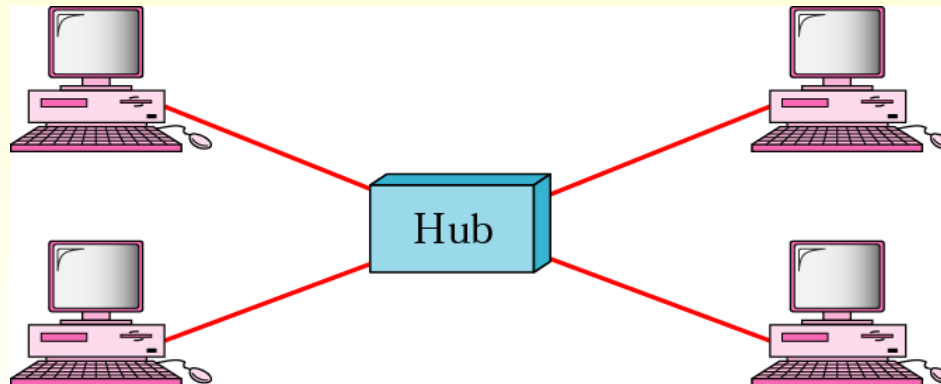
- **Physical Topology (for LANs)**
 - refers to the way in which a network is laid out physically
 - two or more devices connect to a **link**; two or more links form a **topology**
 - four basic topologies are possible: **mesh**, **star**, **bus**, **ring**

■ Mesh

- every device has a dedicated point-to-point link to every other device
- every device must have $n-1$ I/O ports, where n is the number of devices connected
- advantages
 - no traffic problem (no congestion)
 - robust; a failure of a link has no effect on others
 - fault identification and isolation are easy
 - privacy or security (provided there is no wire tapping)
- disadvantages
 - amount of cabling and I/O ports needed (expensive)

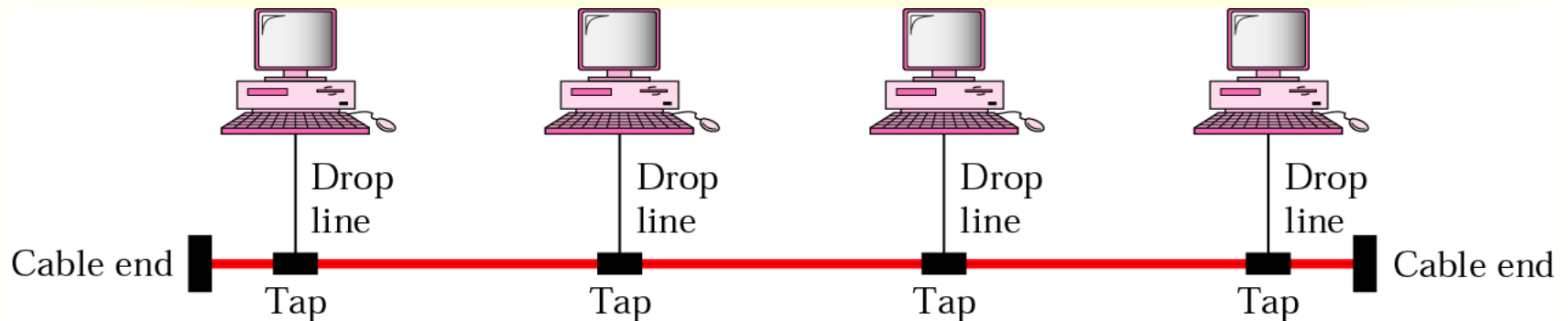


- **Star**
 - each device has a dedicated point-to-point link only to a central controller, usually called a **hub**
 - **advantages**
 - robust; a failure of a link has no effect on others
 - fault identification and isolation are easy
 - less expensive than mesh (but more expensive than others)
 - **disadvantage**
 - failure of the hub halts the system (single point of failure)



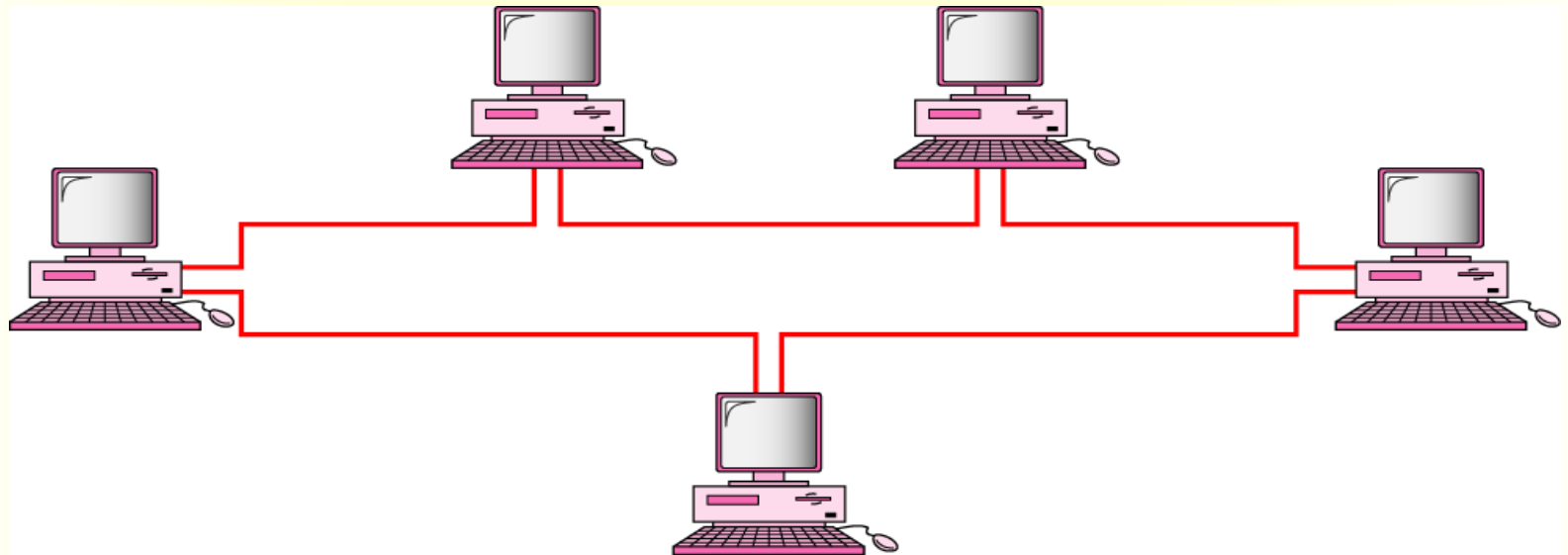
■ Bus

- multipoint (one long cable acts as a backbone to link all the devices in the network)
- advantages
 - ease of installation; less cabling than star or mesh
- disadvantages
 - signal reflection at the taps can cause degradation in quality (solution: limit the number and spacing of devices connected to a given length of cable)
 - difficult reconnection (adding new devices) and fault isolation
 - a fault in the bus cable stops all transmission



■ Ring

- each device has a dedicated point-to-point connection only with the two devices on either side of it
- a signal is passed along the ring in one direction, from device to device, until it reaches its destination
- each device incorporates a repeater (to regenerate bits received before passing it)



- **advantages**
 - **relatively easy to install and configure - adding or deleting a device requires changing only two connections**
 - **fault isolation is simplified (if one device does not receive a signal within a specified period, it can issue an alarm)**
- **disadvantages**
 - **a break in the ring (such as a disabled station) can disable the entire network**

- **Network Categories:** based on size, ownership, the distance it covers, and its physical architecture
 - **Personal Area Network (PAN):** meant for one person; e.g. a wireless network connecting a computer with its mouse, keyboard and printer
 - **Local Area Network (LAN):** usually privately owned and links devices in a single office, building or campus
 - User rates: 10 Mbps – 1 Gbps
 - e.g., Ethernet, Token ring, Apple-talk
 - **Metropolitan Area Network (MAN):** designed to extend over an entire city; it may be a single network or interconnected LANs
 - **Wide Area Network (WAN):** covering large geographic area; may utilize public, leased, or private communications equipment
 - 1980's: 10 Kbps; 2000's: 2.5 Gbps
 - user access rates: 56 Kbps - 155 Mbps

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

■ The Internet

- when two or more networks are connected, they become an **internetwork**, or **internet**
- the most notable internet is called the **Internet**, a collaboration of more than hundreds of thousands of interconnected networks
- it came into being in 1969 - by **ARPA** (Advanced Research Project Agency) of **DoD** for researchers they funded to share findings
- it developed a **packet-switched** network, named as the **ARPANET**

- by the mid-seventies, a full suite of protocols, called **TCP** were developed and proposed as the standard underlying technology for the expanded use of the ARPANET system
- TCP was later split into two protocols: **IP** to handle datagram routing and **TCP** to be responsible for higher-level functions such as segmentation, reassembly, and error correction; it was named as **TCP/IP** and was adopted as the standard protocol suite in 1983
- addressing used by ARPANET users became easier to handle in 1984 with the development of the Domain Name System (**DNS**); this service took the responsibility of putting ARPANET numerical address into readable text form and is still in use
- in the 1990s, an international consortium under the name of European Center for Nuclear research (**CERN**) created the concept of **web pages** and the **WWW** that allowed the exchange of data on the Internet with ease - Tim Berners-Lee

- **Search Engines:** **ARCHIE** (the first) was introduced at McGill University; followed by **gopher** (1991, University of Michigan), **VERONICA** (1993, McGill); today we have such search engines like **AltaVista**, **Yahoo**, **Google**, etc.
- **HTML** was developed in 1992
- **Web Browsers:** **mosaic** (the first graphical web browser) was developed in 1993 by students at the national center for super computing at the university of Illinois; today we have browsers like **Netscape** (although it is dying against my wish), **Internet Explorer**, and **Firefox**
- **the Internet today**
 - end users use the services of **ISPs**
 - there are **international** ISPs that connect nations, **regional**, and **national** ISPs
 - **NAP** - Network Access Points connect backbone networks of ISPs

old figures

- The number of Internet hosts advertised in the DNS rose from 1,313,000 in January 1993 to 625,226,456 in January 2009, an average annual increase of 150%; in the sixteen-year period between 1993 and 2009, the absolute increase in the number of Internet hosts is 47,618%!

Do you know that 80% of the web sites are in the USA?

- What is the picture in Ethiopia?

1.3 Network Protocols and Standards

- a **protocol** is a set of rules that governs data communications
- a protocol defines **what** is communicated, **how** it is communicated, and **when** it is communicated
- for instance, for one computer to send a message to another computer, the first computer must perform the following general steps
 - break the data into small sections called **packets**
 - add addressing information to the packets identifying the source and destination computers
 - deliver the data to the network interface card for transmission over the network

- the receiving computer must perform the same steps, but in reverse order
 - accept the data from the NIC
 - remove transmitting information that was added by the transmitting computer
 - reassemble the packets of data into the original message
- the key elements of a protocol are syntax, semantics, and timing
 - **syntax**: refers to the structure or format of the data
 - **semantics**: refers to the meaning of each section of bits
 - **timing**: refers to when data should be sent and how fast they can be sent
- functions of protocols
 - each device must perform the same steps the same way so that the data will arrive and reassemble properly; if one device uses a protocol with different steps, the two devices will not be able to communicate with each other

- the following are categories of functions that form the basis of protocols
 - encapsulation
 - segmentation and reassembly
 - connection control
 - addressing
 - multiplexing
 - transmission services

1. Encapsulation

- the inclusion of control information to data is called **encapsulation**. Each frame contains not only data but also control information
 - such control information falls into three categories: **address** (sender/receiver), **error-detecting code** and **protocol control** (information about protocol functions)

2. **Segmentation** (by the sender also called **fragmentation**) and **reassembly** (by the receiver)
- a block of data for transmission is of some bounded size
 - at the application level, we refer to a logical unit of data transfer as a **message**
 - lower level protocols may need to break the data up into blocks of some bounded size
 - this process is called **segmentation**
 - reasons for segmentation:
 - the communication network may only accept blocks of data up to a certain size
 - error control may be more efficient with a smaller frame size; fewer bits need to be retransmitted when a frame gets corrupted
 - facilitates more equitable access to shared transmission facilities (for example, without maximum block size, one station could monopolize a multipoint medium)

- **disadvantages**
 - since a frame contains certain amount of control information, the smaller the block size, the greater the percentage overhead
 - frame arrivals may generate an interrupt that must be serviced; hence smaller blocks result in more interrupts
 - more time is spent processing smaller, more numerous frames
- the counterpart of segmentation is **reassembly**
 - on receipt, the receiving device must reassemble the segmented data into a message appropriate to the application

3. Connection control

- two types of protocols (services)
 - **Driving own car vs taking a taxi or Anbessa Autobus**
 - **connectionless** service
 - the packets are sent from one party to another with no need for connection establishment or connection release
 - the packets are not numbered, they may be delayed, lost, or arrive out of sequence; there is no acknowledgement either
 - **UDP (User Datagram Protocol)**, one of the transport layer protocols, is connectionless
 - good for one time transaction; e.g., email

- **connection-oriented service**
 - in connection-oriented data transfer, a logical association or connection is established between the communicating computers (devices)
 - long sustained session
 - orderly and timely delivery of packets, e.g., FTP
- **TCP (Transmission Control Protocol)** is connection-oriented
- three phases are involved
 - connection establishment (agreement to exchange data)
 - data transfer (data and control information exchanged)
 - connection termination (termination request) - by any of the two parties
- the key characteristics of connection-oriented data transfer is that sequencing is used
- each side sequentially numbers the frames that it sends to the other side

- because each side remembers that it is engaged in a logical connection, it can keep track of both outgoing numbers, which it generates, and incoming numbers which are generated by the other side
- sequencing supports three main functions
 - **ordered delivery**: frames may not arrive in the order in which they were sent, because they may traverse different paths
 - **flow control**: a receiving station has to limit the **amount** or **rate** of data that is sent by a transmitting station
 - the simplest form of flow control is **stop-and-wait** procedure in which each frame must be acknowledged before the next can be sent
 - more efficient protocols involve some form of credit provided to the transmitter, which is the amount of frames that can be transmitted without an acknowledgement; e.g. **sliding window** technique

- **error control**: error control is implemented as two separate functions; error detection and retransmission
 - if an error is detected, the receiver discards the frame
 - upon failing to receive an acknowledgement to the frame in a specified reasonable time, the sender retransmits the frame
 - some protocols also employ error correction which enables the receiver not only to detect errors but, in some cases, to correct them
- *we will study some details of flow control and error control again in the data link layer and in TCP*

4. **addressing**

- a unique address is associated with each end-system in a configuration (e.g. workstation, server) and each intermediate system (e.g. router)
- an example is an IP address in TCP/IP connection, but addressing spans more layers: data link, network, transport

5. **multiplexing**

- occurs when multiple connections share a single connection (multiple access)

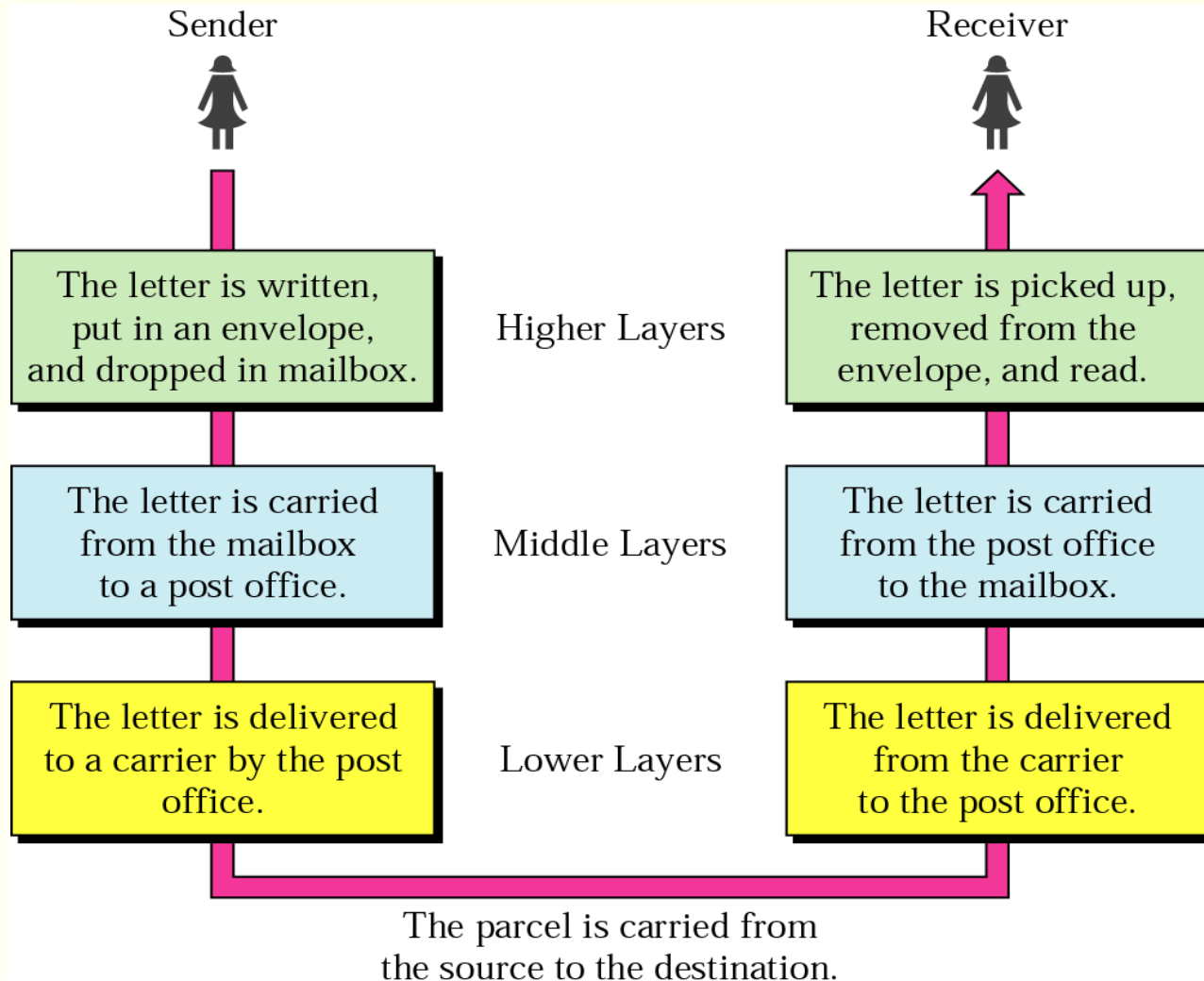
6. **transmission services**

- a variety of additional services can be provided
- **priority**: messages such as control messages may need to get through to the destination station with minimum delay
- **security**: security mechanisms, restricting access, may be invoked

- **Protocols in a layered architecture**
 - protocols that work together to provide a layer or layers of the model are known as a **protocol stack** or **suite**, e.g. TCP/IP
 - each layer handles a different part of the communications process and has its own protocol
- **Data Communication Standards**
 - standards are essential for **interoperability**
 - data communication standards fall into two categories
 - **De facto standards**: that have not been approved by an organized body; mostly set by manufacturers
 - **De jure standards**: those legislated by an officially recognized body such as ISO, ITU, ANSI, IEEE

1.4 Network (Reference) Models

- Consider an example of sending a letter



- **Hierarchy**

- there are three different activities at the sender side and another three at the receiver's side
- the tasks must be done in the order given in the hierarchy

- **Layers and Services**

- within a single machine, each layer uses the services immediately below it and provides services for the layer immediately above it
- between machines, layer x on one machine communicates with layer x on another machine

- **Why layering?**
 - **reduces complexity (one big problem to smaller ones)**
 - **standardizes interfaces (between layers)**
 - **facilitates modular engineering (different teams work on different modules)**
 - **assures interoperable technology**
 - **accelerates evolution of networking technology**
 - **simplifies teaching and learning**

- Two important network models or architectures
 - The ISO OSI (Open Systems Interconnection) Reference Model
 - The TCP/IP Reference Model

a. The OSI Reference Model

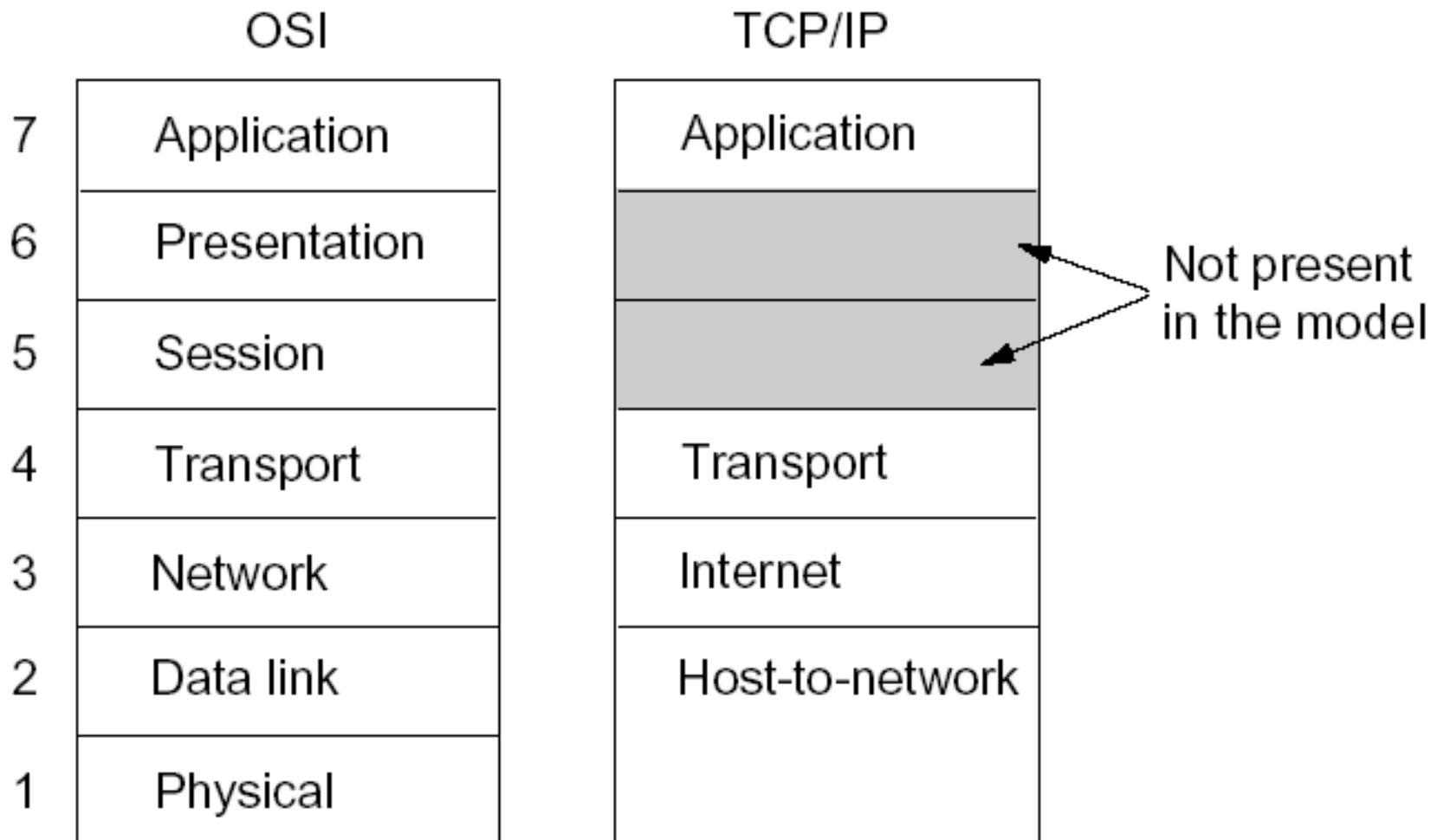
- consists of 7 layers
- was never fully implemented as a protocol stack, but a good theoretical model
- **Open** – to connect open systems or systems that are open for communication with other systems

- **Principles to arrive at the 7 layers**
 - **a layer should be created at different level of abstraction**
 - **a layer should perform a well-defined function**
 - **the function of each layer should be chosen towards defining internationally standardized protocols**
 - **layer boundaries should be chosen to minimize the information flow across the interfaces**
 - **the number of layers should be optimal**
 - **large enough so that distinct functions should not be put together, and**
 - **small enough so that the architecture does not become unwieldy**

b. The TCP/IP Reference Model

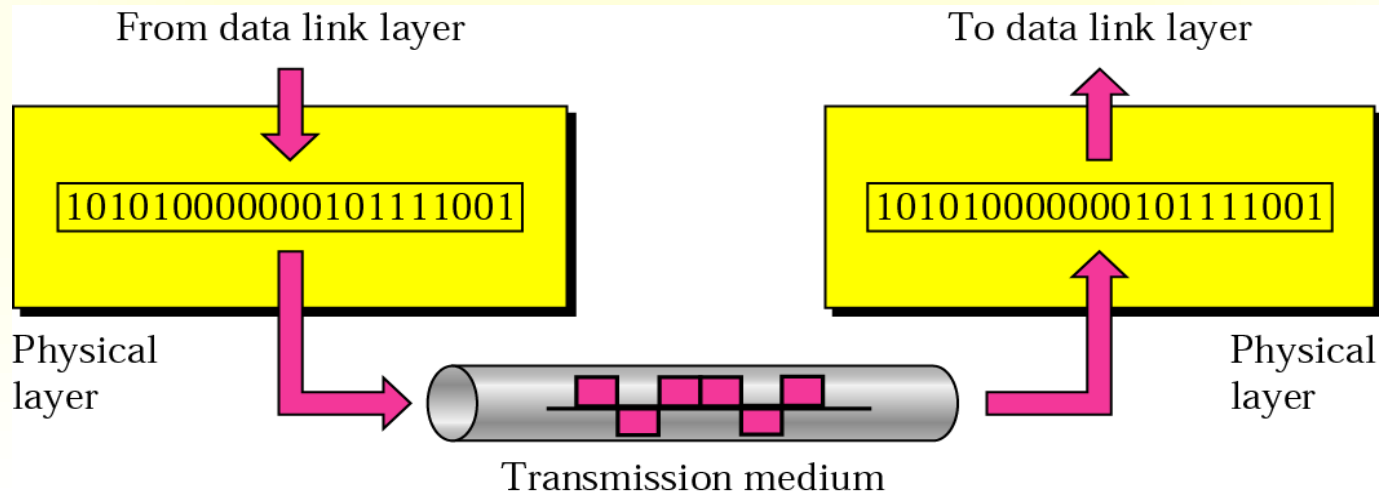
- **TCP/IP - Transmission Control Protocol/Internet Protocol**
- **used by ARPANET and its successor the Internet**
- **design goals**
 - **the ability to connect multiple networks (internetworking) in a seamless way**
 - **the network should be able to survive loss of subnet hardware, i.e., the connection must remain intact as long as the source and destination machines are properly functioning**
 - **flexible architecture to accommodate requirements of different applications - ranging from transferring files to real-time speech transmission**
- **these requirements led to the choice of a packet-switching network based on a connectionless internetwork layer**
- **has 4 (or 5 depending on how you see it) layers: Application, Transport, Internet (Internetwork), Host-to-network (some split it into Physical and Data Link)**

■ OSI and TCP/IP Layers Correspondence



1. Physical Layer

- responsible for transmitting individual bits from one node to the next
- design issues:
 - **mechanical**: the size and shape of the network connector, how many pins does the network connector has and what each pin is used for
 - **electrical**: how many volts represent a 1 and how many a 0
 - **timing**: how many nanoseconds a bit lasts
 - whether communication is one way or in both directions simultaneously



2. Data Link Layer

- makes the line appear free of undetected transmission errors to the network layer (flow and error control)
- this layer was not required if the underlying transmission media were error free
- how?
 - the sender breaks the input data into **data frames** and transmit them sequentially (numbering them) by adding error detecting codes
 - the receiver confirms correct receipt by sending back an **acknowledgement frame**
- note: the Host-to-network layer in the TCP/IP model is not defined; it simply points out that the host has to connect to the network using some protocol so that it can send IP packets to it

3. **Network Layer (or Internet Layer)**

- **permits hosts to inject packets into any network and have them travel independently to their destination - possibly on a different network**
- **allows heterogeneous networks to communicate - addressing, packet size, protocol differences**
- **arrival order of packets may not be respected - let higher layers do the rearrangement if need be**
- **similar to the snail mail system - an international mail from country A to country B traveling through one or more international mail gateways transparent to the user and other hidden rules such as stamps, preferred envelope size, delivery rules, ...**

- **decides how packets are routed from source to destination**
 - **three mechanisms**
 - **based on static tables,**
 - **can be determined at the start of each conversation,**
 - **can be highly dynamic by deciding a new route for each packet**
- **controls congestion**
- **TCP/IP defines an official packet format and protocol called IP for delivering IP packets and routing**
- **not essential for LANs**

4. **Transport Layer**

- **accepts data from above, splits it up into smaller units if need be, passes them to the network layer, and ensures that the pieces all arrive correctly at the other end**
- **allows peer entities on the source and destination machines to hold conversations**
- **in the case of the OSI model, determines types of services to provide to the session layer**
 - **deliver messages in the order they were sent **or** without guarantee of the order of delivery**
 - **broadcasting messages to multiple destinations**

- **In the TCP/IP model, two end-to-end transport protocols have been defined**
 - i. **TCP - Transmission Control Protocol**
 - **a reliable connection-oriented protocol that allows a byte stream to be delivered without error**
 - **handles flow control to make sure that a fast sender does not swamp a slow receiver**
 - ii. **UDP - User Datagram Protocol**
 - **an unreliable, connectionless protocol**
 - **for applications**
 - **that do not want TCP's sequencing or flow control and wish to provide their own**
 - **where prompt delivery is more important than accurate delivery, e.g., audio and video**

5. **Session Layer** (OSI model)

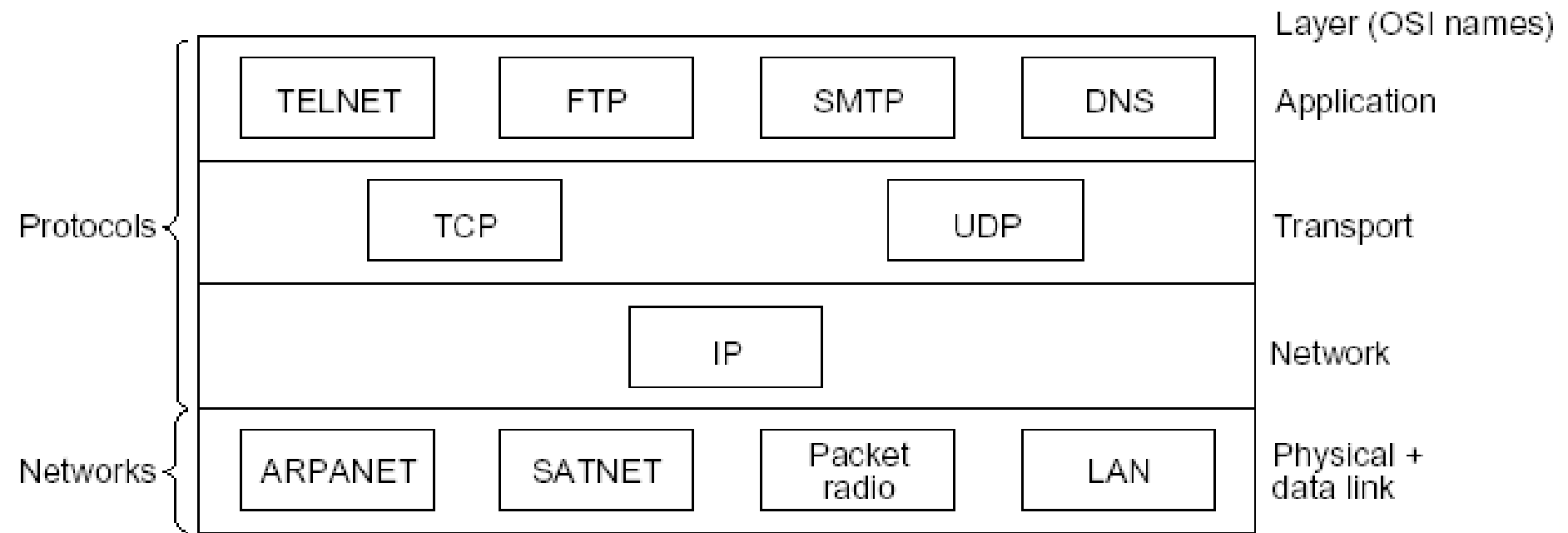
- allows users to establish sessions between them
- services include
 - dialog control - keeping track of whose turn it is to transmit
 - token management - preventing two parties from attempting the same critical operation at the same time
 - synchronization - checkpointing long transmissions to allow them to continue from where they were after a crash
 - provides access rights, billing functions, etc.

6. **Presentation Layer** (OSI model)

- concerned with the syntax and semantics of the information transmitted
- different data structures can be defined in an abstract way so that it is possible for computers with different data representations to communicate
- provides character code conversion, data encryption, data compression

7. Application Layer

- **contains all the higher level protocols that are commonly needed by users; examples are**
 - **HTTP (Hypertext Transfer Protocol) - for fetching pages on the WWW**
 - **TELNET - virtual terminal, to log on to a remote machine**
 - **FTP - file transfer**
 - **SMTP - e-mail**
 - **DNS - for mapping host names onto their network addresses**



Protocols and Networks in the TCP/IP model initially

Host (upper) Layers

Summary

- **Application (7):** Network services to applications
- **Presentation (6):** Data presentation to applications
- **Session (5):** Managing sessions between applications
- **Transport (4):** End-to-end connection and reliability
- **Network (3):** Managing connections across the network
- **Data Link (2):** Reliable data delivery across the link
- **Physical (1):** Physical characteristics of the media

Media (lower) Layers

- **Layers involved in various hosts (TCP/IP)**
 - when a message is sent from device A to device B, it may pass through many intermediate nodes
 - the intermediate nodes usually involve the first three layers

